Holistic Service Provisioning in a UAV-UGV Integrated Network for Last-Mile Delivery

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***Abstract*—Effective last-mile delivery is pivotal in smart logistics system. While existing delivery network architectures, such as Drone-as-a-Service (DaaS), are capable of enhancing order delivery effectiveness, they often fall short in provisioning diverse delivery services. Furthermore, DaaS-based last-mile delivery systems face challenges from limited payload capacity and range. In this paper, we propose a UAV-UGV integrated network architecture based on Multi-access Edge Computing (MEC), denoted as DaaS^{+}, encompassing the diverse delivery services from both Unmanned Aerial Vehicle (UAV) and Unmanned Ground Vehicle (UGV) for last-mile delivery. To optimize the effectiveness in the delivery process, the intricate overhead and constraints of heterogeneous delivery services are taken into the full consideration. Specifically, we present an energy-aware service model for the UAV-UGV integrated network that considers the deadline constraints of services. Additionally, we address issues of service unavailability during service provisioning. To identify optimal service provisioning plans, we design a novel Energy-Aware Holistic Service Provisioning Mechanism based on Particle Swarm Optimization (ES-PSO), which minimizes delivery energy consumption while adhering to service deadline constraints. Experimental results substantiate the effectiveness of our proposed solution, demonstrating its ability to generate superior service provisioning plans and significantly reduce total delivery energy consumption.**

***Index Terms*—Service provisioning, drone-as-a-service, multi-access edge computing, last-mile delivery.**

I. Introduction

A. Motivation

LAST-MILE delivery is a crucial aspect of the modern smart logistics system [1], [2]. According to Statista, global retail e-commerce sales are forecast to reach around \$8.1 trillion in 2026 and are expected to continue growing [3]. The rise of e-commerce contributes significantly to the demand for last-mile delivery service [4].

The core factor of last-mile delivery relies on Unmanned Aerial Vehicles (UAVs) or Unmanned Ground Vehicles (UGVs) with the support of various cloud services to achieve automation and intelligence in last-mile delivery services [5]. Due to the diverse characteristics of delivery service resources and the dynamic delivery environment, it is a great challenge to effectively generate delivery service provisioning plans. Despite the fact that centralized cloud servers contain massive computing resources, they are able to generate the optimal delivery service provisioning plan for orders quickly. The geographical separation between the cloud data center and UAV/UGV frequently leads to significant data transmission delays, and these delays significantly impact the real-time delivery of orders [6]. In contrast, Multi-access Edge Computing (MEC) provides a solution by deploying resources and services at the network edge. It is capable of providing real-time, distributed and highly reliable service support for delivery UAVs or UGVs. However, the dynamics and diversity of MEC environments also have the potential to influence the effectiveness of the delivery system [7], [8]. Therefore, effectiveness service provisioning under the Quality of Service (QoS) requirement for MEC-based last-mile delivery systems is a critical issue.

The emergence of Drone-as-a-Service (DaaS) has introduced a novel service paradigm for modeling UAV-based delivery services. This paradigm enables the implementation of large-scale delivery mechanisms and facilitates the collaboration among different UAVs in carrying out long-distance delivery tasks [9]. In contrast, other delivery services such as UGVs demonstrate greater resilience to dynamic weather conditions, and offer several advantages including reduced safety risks and noise levels [10]. How to ensure the separate advantages of UAVs and UGVs in the last-mile delivery, and achieve the collaboration delivery of UAVs and UGVs is one of the most critical issues. Current DaaS architecture does not support the collaboration among heterogeneous delivery services such as UAVs and UGVs. To fill such a gap, it is imperative to develop new UAV-UGV integrated network architecture (DaaS^{+}) based on MEC and service management mechanisms to optimize the performance of last-mile delivery systems.

Fig. 1 illustrates an example UAV-UGV integrated network based on Antworks's ADNET project[1] [11], [12]. A typical

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[1]https://www.antwork.link/

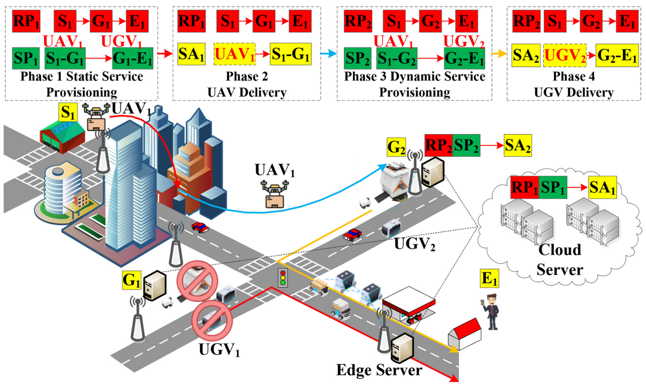


Fig. 1. An example of UAV-UGV integrated network for last-mile delivery.

delivery process consists of four major phases. Specifically, the first phase is the static service provisioning phase where route planning (RP_1) and a service provisioning plan (SP_1) is generated in cloud servers. The second phase is the UAV delivery phase where the UAV_1 transports the cargo to the designated delivery stations (G_1) according to the service allocation plan (SA_1). The third phase is the dynamic service provisioning phase where the availability of the target delivery stations (G_1) is being constantly monitored during the UAV flight. If the station becomes unavailable, a rapid local path planning is conducted to devise an alternative delivery plan $(RP_2,\ SP_2)$ in edge servers. Finally, in the UGV delivery phase, the UGV delivers the cargo to its final destination (G_2) and completes the entire delivery process according to the service allocation plan (SA_2).

The integration of UAV and UGV delivery services presents several significant challenges that must be addressed to optimize the performance of last-mile delivery systems. Below, we analyze the primary challenges and discuss how this paper addresses them.

1) Heterogeneous Service Collaboration: One of the foremost challenges is to ensure the seamless collaboration between UAVs and UGVs, given their distinct operational characteristics and advantages. UAVs offer rapid delivery capabilities but are limited by factors such as battery life and susceptibility to weather conditions. In contrast, UGVs provide more stable and reliable services but at a slower pace. Coordinating these heterogeneous services to maximize their respective strengths while minimizing their weaknesses is a complex task. This paper proposes a novel UAV-UGV integrated network architecture based on MEC, which facilitates the dynamic coordination and optimization of these services, ensuring efficient and reliable last-mile delivery.

2) Dynamic Service Provisioning: Another significant challenge is to manage the dynamic nature of delivery environments. This includes handling service unavailability due to various factors such as weather conditions, traffic, and unforeseeable obstacles. The proposed dynamic service provisioning phase constantly monitors the availability of delivery stations and employs rapid local path planning and alternative delivery strategies to address service disruptions. By incorporating real-time data and edge computing capabilities, our approach ensures that delivery services can adapt quickly to changes, maintaining service reliability and efficiency.

B. Contributions

To utilize the advantages of UAVs and UGVs delivery, a UAV-UGV integrated network architecture is proposed. It consists of UAV and UGV delivery services. Besides, the problems of service provisioning are also taken into consideration. The aim of this paper is to propose a novel holistic service provisioning mechanism for UAV-UGV integrated network with the major optimization objective of minimizing the total delivery energy consumption. The main contributions of this paper are as follows:

1) A UAV-UGV integrated network architecture based on MEC (named DaaS$^+$) is designed. This architecture serves as an extension of the existing DaaS. It aims to significantly enhance the synergy and collaboration among a diverse array of delivery services, addressing the shortcomings prevalent in conventional networks. The integration of UAVs and UGVs within the architecture signifies a pioneering step towards creating a comprehensive and seamless delivery network.

2) An energy-aware service model specifically tailored for the UAV-UGV integrated network is presented. This model takes into account the deadline constraints of services during the delivery process. By considering the intricacies and constraints of heterogeneous delivery services, the proposed model aims to optimize the effectiveness of delivery using UAVs and UGVs. It signifies a notable advancement in addressing the energy consumption challenges within the delivery network while ensuring adherence to service deadline constraints.

3) To overcome issues of service unavailability during service provisioning and to identify optimal service provisioning plans, an Energy-aware holistic Service provisioning mechanism based on Particle Swarm Optimization is proposed, referred to as ES-PSO. ES-PSO is employed to identify the most effective service provisioning plan, with the primary objective of minimizing energy consumption during the delivery process. The ES-PSO consists of two integral phases: the static and dynamic stages. In the static stage, the ES-PSO is utilized to minimize the total delivery energy consumption while adhering to deadline constraints. In the dynamic stage, ES-PSO formulates a plan for re-selecting the UAV-UGV integrated network services. This plan is instrumental in detecting and resolving service unavailable during the delivery process.

4) Simulation experiments were conducted utilizing the intelligent UAV-UGV integrated network known as EXPRESS 2.0, accompanied by a dataset representing a real-world delivery environment. The experiments demonstrate the superiority of the presented approach in generating optimized service provisioning plans and notably reducing the overall energy consumption in the delivery network. We further enhance and validate the effectiveness of the ES-PSO mechanism by adjusting the penalty coefficient and performing an ablation study. The simulation results substantiate the effectiveness of the proposed mechanism.

The rest of this paper is organized as follows: Section II provides an overview of the related work. Section III presents the novel DaaS^{+} UAV-UGV integrated network architecture based on MEC and formulates the service model and service provisioning problem. Section IV proposes holistic service provisioning mechanism for UAV-UGV integrated network. Section V conducts comprehensive simulation experiments based on real-world map data to evaluate the performance of proposed mechanism. Finally, in Section VI, we conclude the paper and point out some future research directions.

II. Related Work

A. UAV-Enabled Last-Mile Delivery Systems

UAV-enabled last-mile delivery systems represent a transformative advancement in the logistics industry, with real-world examples showcasing their potential [13], [14], [19]. Companies like Amazon have pioneered the use of drones to deliver packages to customers' doorsteps in record time [20]. Similarly, companies such as Zipline have revolutionized medical supply deliveries in remote or disaster-stricken areas. They utilize UAVs to expedite the transportation of vital medications and supplies to locations that are otherwise challenging to access [21]. These examples underscore how UAV-enabled last-mile delivery systems transcend conventional boundaries, providing rapid and effective solutions that address pressing challenges across industries, from e-commerce giants to humanitarian efforts.

B. Service Management for MEC

A significant number of studies focuses on service management in edge computing [22], [23], [24]. For example, Li et al. proposed a service provisioning framework and designed an effective service provisioning cost minimization algorithm based on the framework to address the service provisioning problem [25]. Similarly, Wang et al. proposed a green service composition method which achieves green service composition optimization by minimizing the energy and network resource consumption of physical servers and switches in cloud data centers [26]. While numerous studies focus on service composition in edge computing environments, it is worth noting that generic service management approaches often exhibit reduced effectiveness when dealing with services in smart logistics systems. These services are notably more diverse and complex. Siasi et al. introduced novel "on-demand" heuristic solutions and batch sorting methodologies to address service survivability in fog computing domains, taking into account end-user and operator concerns such as service delay, bandwidth, and resource parameters. It aims to support clients with stringent failure recovery requirements by optimizing SFC service mapping and protection [27].

C. Service Provisioning for UAV/UGV Delivery Systems

Given the increasing popularity of UAVs and UGVs, several studies have focused on service management in smart logistics systems. She et al. used UAVs for last-mile delivery and proposed a plan to solve the shortest path of UAVs, taking into account constraints such as UAV battery capacity and delivery distance, to reduce the operating cost and energy consumption of UAVs [15]. Liang et al. proposed a CMfg-SC for QoS-aware DRL involving logistics. They have designed a dueling Deep Q-Network (DQN) with prioritized replay named PD-DQN [16]. Liu et al. introduces a parallel automatic delivery service model for last-mile superchilling product delivery using logistics drones. It optimized social value, energy efficiency, and productivity, and validated through simulation experiments in urban environments [13].

While only a few studies have considered service collaboration in MEC-based smart logistics systems, Chu et al. proposed a multi-drone collaboration approach for the last-mile delivery problem and introduced a service provisioning optimization algorithm to improve the effectiveness of multi-drone delivery [17]. Bacheti et al. used a formation of UAVs and UGVs to complete the delivery task, with the UAVs returning to the UGVs after completing the delivery task to continue the delivery task on the UGV's route [18]. Xia et al. proposed a novel two-layer path planning method for cooperative ground vehicle (GV) and drone systems, optimizing area coverage and travel paths. Simulations in Changsha, China, and random instances validated method's effectiveness [28]. However, the existing work failed to take full advantage of UAVs and UGVs so as to realize the best collaboration of multiple delivery services.

D. Discussions

The literature review presented in Table I highlights that existing research on the service provisioning problem for UAV/UGV delivery systems has predominantly focused on either UAV service provisioning or UGV service provisioning, with limited attention to both UAV and UGV service provisioning. Furthermore, most existing studies fail to consider the unavailability of services during the delivery process and lack real-world scenario demonstrations. Consequently, they do not fully leverage the unique advantages and characteristics of UAVs and UGVs to minimize delivery energy consumption while ensuring deadline constraints.

Hence, the objective of our research is to develop an effectiveness UAV-UGV integrated network architecture known as DaaS^{+}, and introduce an innovative holistic service provisioning mechanism tailored for this integrated network architecture.

III. System Architecture and Problem Formulation

A. System Architecture

Current DaaS architectures do not facilitate collaboration among different types of delivery vehicles, UAVs and UGVs. This limitation prevents the integration of their separate advantages and hinders the potential for collaborative last-mile delivery. A UAV-UGV integrated network based on MEC improves order delivery effectiveness by collaborating with various delivery services. Nonetheless, the diverse attributes of delivery services pose challenges for effective service collaboration. Therefore, an generalized integrated UAV-UGV

TABLE I
RELATED WORKS AND THEIR PROPERTIES

Reference	Objective	UAV Service	UGV Service	Service Unavailability	Real-world Demonstration
[13], [14]	Energy consumption	✓	×	×	×
[15]	Cost, Energy consumption	✓	×	×	×
[16]	Energy consumption	×	×	×	×
[17]	Energy consumption, Deadline	✓	×	×	×
[18]	Time	✓	✓	×	×
This paper	Energy consumption, Deadline	✓	✓	✓	✓

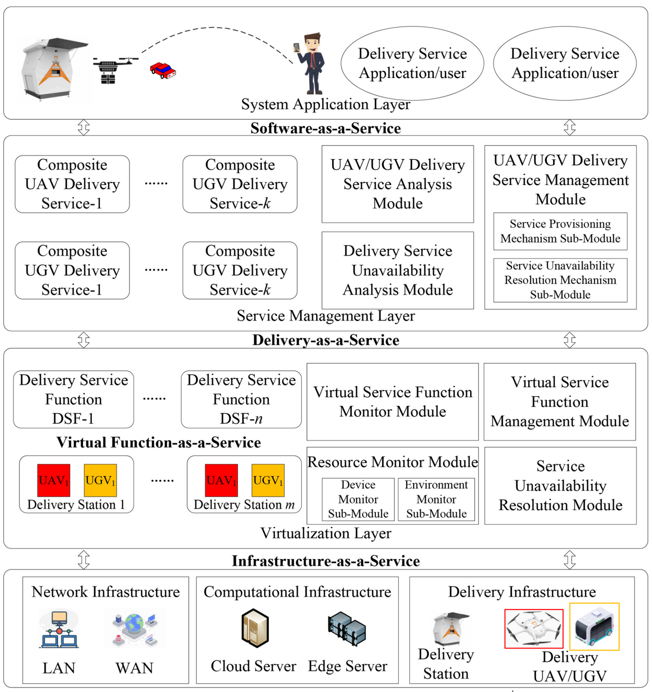


Fig. 2. System architecture of DaaS$^+$.

network architecture is necessary for providing effective service collaboration with the least amount of energy and time consumption.

As illustrated in Fig. 2, the system architecture of DaaS$^+$ consists of four distinct layers: the system application layer, service management layer, virtualization layer, and infrastructure layer. The system application layer is responsible for interacting with end-users and managing delivery applications/users. This layer includes various delivery service applications and user service requests that facilitate order placement, tracking, and other customer-related activities.

The service management layer serves as the focal point for generating an optimal service provisioning plan and effectively resolving service resource unavailable. It comprises essential components including the UAV and UGV delivery service analysis module. The delivery service unavailability analysis module plays a crucial role in identifying and resolving potential unavailability in service allocation. The delivery service management module is responsible for overseeing and coordinating the entire spectrum of delivery services. This module contains two sub-modules: the service provisioning mechanism sub-module and the service unavailability resolution mechanism sub-module. These sub-modules synergistically contribute to generate delivery service provisioning plans and resolution plans based on optimization objectives.

The virtualization layer establishes a unified abstraction of the diverse resources within the delivery infrastructure, harmonizing their heterogeneity. The virtualization layer houses several crucial modules. The virtual service function monitor module plays an active role in overseeing the execution of virtual delivery service functions. The module for resource monitor continually tracks and manages the allocation and utilization of the underlying resources for delivery UAVs/UGVs. The virtual service function management module is responsible for the deployment, scaling, and effective management of virtualized service functions to ensure the implementation of the service provisioning plan by the service management layer. The service unavailability resolution module dynamically identifies and resolves potential unavailable stemming from resource contention or overlapping service demands, ensuring seamless and uninterrupted service delivery.

The infrastructure layer forms the foundation of the MEC-based UAV-UGV integrated network. It encompasses the physical components, including cloud/edge computing servers, communication network, UAVs, and UGVs. This layer is responsible for providing the necessary hardware resources and connectivity to enable seamless communication between the various components of the system.

The workflow of DaaS$^+$ begins with user engagement through the application layer, facilitating delivery order placement and tracking. The service management layer assesses UAV and UGV capabilities, resolves unavailability, and coordinates services. Central to this layer is the delivery service management module, orchestrating optimal service provisioning plan and unavailability resolution. The virtualization layer abstracts diverse delivery resources, providing virtualized delivery assets via DaaS$^+$. It encompasses modules for function monitoring, function management, resource monitor, and service unavailability resolution. With effective resource utilization ensured, the system executes delivery tasks using UAVs and UGVs. Continuous monitoring and user feedback inform ongoing refinements, closing the loop and enhancing MEC-based UAV-UGV integrated network effectiveness.

TABLE II
DEFINITION OF NOTATIONS

Notation	Definition
K/D_k	Number/set of delivery service
I/R_i	Number/set of delivery order request
C/RS_c	Number/set of service unavailability resolution plan
$D_A(N_S, N_D)$	Minimum UAV route distance between N_S and N_D
$D_G(N_S, N_D)$	Minimum UGV route distance between N_S and N_D
W	Weight of delivery cargo
P_{opt}	Optimal delivery service provisioning plan
P_{res}	Optimal service unavailability resolution plan

B. Service Models

In this section, two types of service models are defined and described for UAV-UGV integrated network: DaaS$^+$ static and dynamic service models, corresponding to the static and dynamic phases of the UAV-UGV integrated network. The static service model is primarily used to solve the delivery service provisioning problem before the order delivery stage. This model considers various factors to optimize the delivery service plan, such as service requirements, resource availability and energy constraints. By employing this model, we ensure effective provisioning of delivery services, minimizing energy consumption, and maximizing resource utilization. The dynamic service model is used to solve the service unavailable phenomenon. By employing this model, we ensure that delivery services are executed smoothly and effectively, even when confronted with unexpected changes in the environment or user requirements. The notations are list in Table II.

1) DaaS$^+$ Static Service Model: Fig. 3 demonstrates that the DaaS$^+$ Service Model for UAV-UGV integrated network. The model consists of transport nodes and transport paths. Among them, the transport nodes are categorized into three distinct types: UAV delivery station nodes, UGV delivery station nodes, and receiver nodes. The transport paths are classified into two types: UAV paths (indicated by red lines) and UGV paths (indicated by blue lines). A complete delivery path consists of two parts. Initially, cargoes are dispatched from the start nodes, namely, the UAV delivery station nodes. The shortest flight path between the UAV delivery station nodes is used as the flight route of the UAVs, and it is assumed that it follows the regulations such as a no-fly zone. Secondly, after reaching the relay UGV delivery station nodes, it is transferred to the UGV to continue the delivery, and the UGV delivers the cargo to the receivers through ground roads.

Definition 1: DaaS$^+$ Service Model. DaaS$^+$ is defined as a tuple of $\langle ID, A_{i,j}, G_{m,n}, QoS_A, QoS_G, z_t\rangle$ where;

ID is a unique identifier for a UAV/UGV delivery service;

$A_{i,j}$ is the *j*-th delivery UAV at the *i*-th delivery station;

$G_{m,n}$ is the *n*-th delivery UGV at the *m*-th delivery station;

QoS_A, QoS_G consist of a tuple $\langle q_1, q_2, q_3, \ldots, q_k\rangle$, where each q_k denotes a QoS attribute of UAV/UGV delivery service;

z_t is a binary variable $z_t \in \{0, 1\}$;

Definition 2: DaaS$^+$ QoS Model. The QoS model (QoS_A, QoS_G) of UAV and UGV delivery service is defined as a tuple of $\langle L_{max}, P_{max}, P_{min}, P_D, V\rangle$ where;

$$P_D = \frac{W}{L_{max}} \times (P_{max} - P_{min}) + P_{min} \tag{1}$$

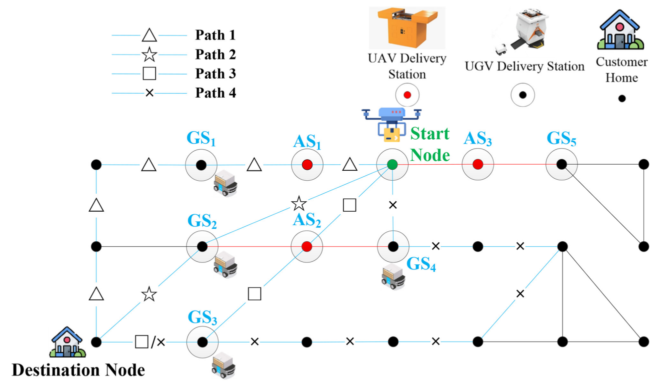


Fig. 3. DaaS$^+$ service model.

L_{max} is UAV/UGV's maximum payload weight capacity. The service provisioning and service unavailability resolution algorithm uses it to achieve the weight-limit DaaS$^+$ selection.

P_{max}, P_{min} are UAV/UGV's maximum and minimum running power. The service provisioning and service unavailability resolution algorithm uses P_{max}, P_{min} as the heuristic to find the energy-aware selection.

P_D is the delivery power during UAV/UGV delivery process. It is calculated based on the weight of delivery cargo, UAV/UGV's maximum payload weight capacity, UAV/UGV's maximum and minimum running power.

V is the UAV/UGV's average delivery speed. The service provisioning and service unavailability resolution algorithm uses it to find the optimal delivery route.

Definition 3: Delivery Order Request. A Delivery Order Request (*DOR*) is defined as a service request to deliver an order from the start station node to the destination station node. ***DOR*** is defined as a tuple of $R_i = \langle W, N_S, N_D, D\rangle$ where;

W is the weight of delivery cargo.

N_S, N_D are the start station node and destination station node, respectively.

D is the deadline constraint of delivery order request.

In the static service stage, the time consumptions of the UAV/UGV delivery phase are T_A and T_G, respectively.

$$T_A = \frac{D_A(N_S, N_D)}{V} \tag{2}$$

$$T_G = \frac{D_G(N_S, N_D)}{V} \tag{3}$$

The energy consumptions of the UAV/UGV delivery phase are denoted as E_A and E_G, respectively.

$$E_A = P_D \times T_A \tag{4}$$

$$E_G = P_D \times T_G \tag{5}$$

2) DaaS$^+$ Dynamic Service Model: In practical delivery operations, during the simultaneous delivery of multiple orders, a scenario may arise where previously allocated UGV services become unavailable due to the necessity of transferring goods between UAVs and UGVs. This occurrence necessitates a resolution of the UGV delivery service for the affected delivery order.

Definition 4: Service Resolution Plan Model. The Service resolution plan is defined as a tuple of $RS_c =< R_i, RG_{m,n}, QoS_G >$, where;

$RG_{m,n}$ denotes the n-th resolution delivery UGV in the m-th delivery station for unavailable service handling. The waiting time required to resume the UGV service at the service unavailability station is denoted as T_{wait}. Therefore, when service unavailable is occurred, the delivery energy consumption ER_{sum} and time TR_{sum} after service resolution are denoted as:

$$TR_{sum} = \begin{cases} min\{T_A + T_G + T_{wait}, T_A + T_{RD}\}; \\ \qquad\qquad T_{sum} + T_{wait} \leq T_{deadline} \\ T_A + T_G; T_{sum} + T_{wait} > T_{deadline} \end{cases} \tag{6}$$

$$ER_{sum} = \begin{cases} min\{E_{sum}, E_A + E_R\}; \\ \qquad\qquad T_{sum} + T_{wait} \leq T_{deadline} \\ E_A + E_R; T_{sum} + T_{wait} > T_{deadline} \end{cases} \tag{7}$$

$$E_R = P_A \times T_{A,R} + P_G \times T_{G,R} \tag{8}$$

$$T_{RD} = T_{A,R} + T_{G,R} \tag{9}$$

P_A, P_G denote the run-time power of the UAV and the UGV. E_R denotes the energy of delivery order to execute the replanning solution. $T_{A,R}$ denotes the time when the UAV arrives at the new *R-th* transfer station. $T_{G,R}$ denotes the time when the UGV arrives at the destination from the *R-th* new delivery station. T_R denotes the time of delivery order to execute the replanning solution.

In the event of service unavailability, if waiting for service recovery does not risk exceeding the delivery order timeout, a comparison of energy consumption is conducted to decide between waiting for recovery or seeking an service resolution plan. If waiting would lead to the total delivery time surpassing the deadline constraint, the latter of the order is re-planned. An alternative UGV delivery service capable of completing the delivery to the destination is sought. Subsequently, the recalculated delivery time and energy consumption pertinent to UGV's resolution plan are determined.

C. Problem Formulation

By leveraging the proposed service model, we initiate an analysis of the service provisioning problem within a UAV-UGV integrated network for last-mile delivery systems. Then, a series of problem models are defined to formulate the service provisioning problem with optimization objectives.

Definition 5: DaaS$^+$ Service Provisioning Problem.

1) The total delivery time T_{sum} for each delivery request can be formulated as follows:

$$T_{sum} = \begin{cases} T_A + T_G & if\ D_k\ is\ available; \\ TR_{sum} & if\ D_k\ is\ not\ available; \end{cases} \tag{10}$$

2) The total delivery energy consumption E_{sum} can be formulated as follows:

$$E_{sum} = \begin{cases} E_A + E_G & if\ D_k\ is\ available; \\ ER_{sum} & if\ D_k\ is\ not\ available; \end{cases} \tag{11}$$

3) Optimization objective of the delivery service selection problem: *FIUV*. The objective is to optimize energy consumption of UAV and UGV during delivery while adhering to the deadline constraints of the delivery orders.

The Integrated Utility Value of Fitness (FIUV) is designed that considers the deadline constraint of delivery order request and the total delivery energy consumption to evaluate the service provisioning plan.

$$FIUV(R_i, D_k) = \begin{cases} E_{sum}, & T_{sum} \leq D; \\ E_{sum} + \dfrac{T_{sum}}{D} \times P_D, & T_{sum} > D; \end{cases} \tag{12}$$

The FIUV is comprised of three key components: the delivery energy consumption and time of UAV/UGV during the entire delivery process and the penalty coefficient. This penalty coefficient is representative of the rate of time by which delivery orders exceed their deadline constraints. The underlying concept hinges on two distinct scenarios. In the cases where the delivery time remains within the deadline constraint, the FIUV is computed as the aggregate energy consumption associated with the entirety of the delivery service. In such instances, the objective function is centered around the optimization of total delivery energy consumption. Conversely, when the delivery time surpasses the deadline constraint, the formula integrates two discrete elements. The first element encompasses energy consumption, while the second element takes into account the penalty coefficient *P* incurred due to exceeding the delivery order's deadline constraint. In this context, the FIUV is directed towards the optimization of UAV/UGV's energy consumption, given the constraint of meeting delivery order's deadline constraint.

In this paper, the optimization object aims to minimize the total FIUV. Therefore, the optimization problem is formulated as follows:

$$min \sum_{i=1}^{I} FIUV(R_i, D_k) \tag{13}$$

s.t

$$R_i.W \leq QoS_A.L_{max} \wedge R_i.W \leq QoS_G.L_{max}, \forall R_i \in R \tag{14}$$

$$\sum_{i=1}^{K} D_k.z_i \leq 1, \forall D_k \in D \tag{15}$$

Constraint (14) signifies that the maximum payload weight capacity must not be breached. Constraint (15) guarantees that each delivery service is provided for a maximum of one delivery order request.

IV. Mechanism Design

A. Mechanism Overview

In UAV-UGV integrated network based on MEC, the improvement of order delivery effectiveness originates from the service provisioning. In order to provide the optimal service provisioning plan with the least delivery energy consumption and time, it is necessary to design the holistic service provisioning mechanism.

The ES-PSO mechanism comprises three key stages: Population Initialization, Static Stage, and Dynamic Stage. It takes delivery service requests as input and generates the optimal service provisioning plan. This mechanism aims to

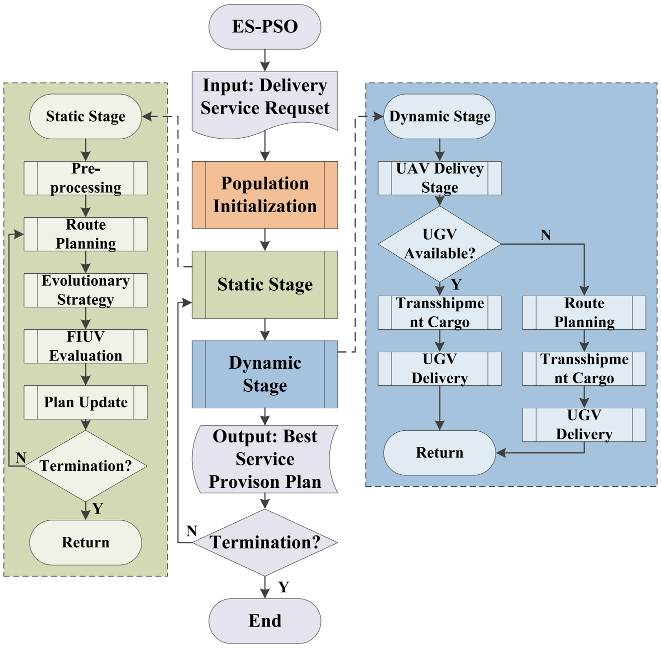


Fig. 4. Process for ES-PSO mechanism.

minimize UAV/UGV's delivery energy consumption under the deadline constraints. The algorithm used in the static stage is Energy-aware Static Delivery Service selection algorithm (denotes as ESDS). The algorithm used in the dynamic stage is Energy-aware Dynamic Service Resolution algorithm (denotes as EDSR). Specifically, the dynamic stage generates the UAV-UGV integrated network service re-selection plan, and used for service unavailable detection and resolution during the delivery process.

The process of ES-PSO mechanism is shown in Fig. 4. During Population Initialization stage, the cloud server receives the delivery service request and sends it to the edge server of the designated warehouse. Then, the edge server requests the service status of local delivery devices and stations. The system filters out the available delivery services and calculates the shortest delivery path between each available delivery station. Based on the above parameters, the initial UAV/UGV service provisioning solutions are generated.

In the static stage, the delivery service request is first analyzed and pre-processed to obtain the characteristics of the cargo to be delivered (weight, size, etc.) and the information about the starting and destination stations of the delivery. Then, the optimal UAV/UAV delivery route plan is generated based on the starting and destination points of the delivery orders. Next, an evolutionary mechanism is used to generate the corresponding UAV/UAV delivery service provisioning plan. Finally, the generated service provisioning plan is evaluated using the FIUV function. Based on the evaluation results, the plan is updated until the termination conditions are met and the optimal service provisioning plan is output.

In the dynamic stage, the UAV starts the delivery process based on the service provisioning plan generated in the static stage. However, the concurrency of multiple orders may cause service unavailability issues. Therefore, when the UAV arrives at the designated transfer station, it checks the current availability of the relay service. If the relay service is available, the order is executed normally according to the static service plan. If the relay service is not available, alternative plans need to be considered. The system considers the following two alternatives:

(1) Waiting for the service to recover at the current station. This option does not increase extra energy consumption. However, it needs to consider order timeouts due to waiting times.

(2) Re-selecting a closer UGV station as the new transfer station. This option may increase the total energy consumption slightly, but it generally does not result in an order timeout. The system compares the time and energy consumption of each plan and finally selects the suitable service re-provision plan.

Algorithm 1 Energy-Aware Static Delivery Service Provisioning Algorithm (ESDS)

Input: Delivery Order Request R, DaaS$^+$ service model diagram D_k, number of individuals N, iteration times *iter*;
Output: P_{opt}; ▷ the best delivery service provisioning plan;
1: **procedure** DCS SUB-ALGORITHM
2: $DS \leftarrow \varnothing$
3: $AList = getListbyDCS(AS_m)$
4: **for** $i = 1$ to $AList.size$ **do**
5: **if** $D_k.QoS_A.L_{max} < R.W$ **then**
6: Remove the i-th UAV from *AList*
7: **end if**
8: **end for**
9: $GList = getListbyDCS(GS_m)$
10: **for** $i = 1$ to $GList.size$ **do**
11: **if** $D_k.QoS_G.L_{max} < R.W$ **then**
12: Remove the i-th UGV from *GList*
13: **end if**
14: **end for**
15: SPSO SUB-ALGORITHM(D_k, A, G, R)
16: **end procedure**
17: **procedure** SPSO SUB-ALGORITHM
18: $Route \leftarrow \varnothing$
19: $GSList = getUseableList(D_k)$
20: $Route \leftarrow getListbyDijkstra(GSList, R)$
21: $ER = getRealPower(R.W, L_{max}, P_{max}, P_{min})$
22: $DS \leftarrow getListbySPSO(A, G, Route, GSList, R)$
23: **return** *DS*
24: **end procedure**

B. ESDS Algorithm

The ESDS algorithm (Algorithm 1) contains two sub-algorithms: the Delivery Carrier Selection algorithm (DCS) and the transfer station Selection based on PSO algorithm (SPSO). The whole process contains five phases. The first phase involves identifying and excluding UAVs and UGVs that do not meet the cargo weight requirement. (Lines 1-15). The second phase is to choose the set of available UGV stations based on the status of the UGV stations (Lines 17-18). The third phase employs the Dijkstra algorithm to determine

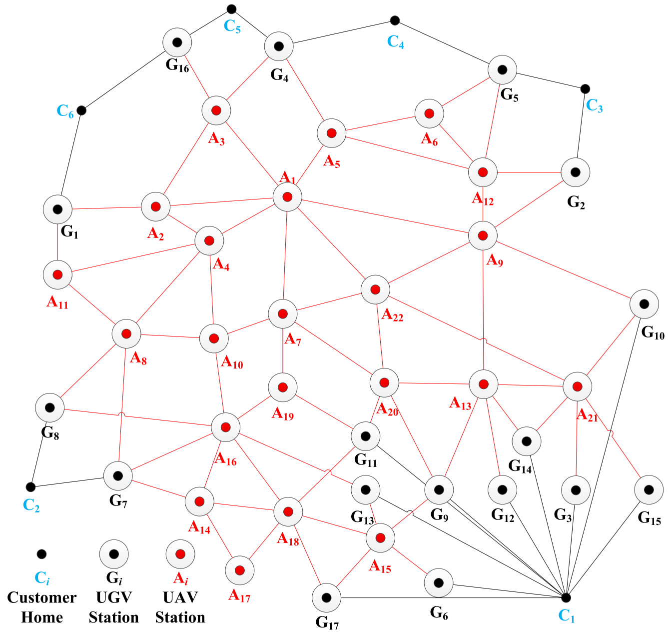


Fig. 5. UAV-UGV integrated network dataset.

the shortest path, generating a set of shortest paths derived from the given origin and destination points. (Line 19). In this regard, the ground path and delivery time of the UGV are calculated by applying the API of AMAP.[2] The fourth phase finds the runtime power of the carrier based on the weight of the cargo using Eq. (1) (Line 20). The fifth phase returns an energy-aware delivery service plan based on the PSO algorithm (Lines 21-22). Since the time complexity of Algorithm 1 mainly comes from using the PSO algorithm to find the delivery service plan, we focus on this part of the analysis. The time complexity of Algorithm 1 is $O(m \times N \times iter)$, where m,N,$iter$ are the number of UGV stations, particles and algorithm iterations.

C. EDSR Algorithm

The EDSR algorithm (Algorithm 2) is to address the problem of sudden unavailability of UGV station services during the delivery process. The algorithm process is as follows: Firstly, the UAV delivers the cargo to the assigned transfer station and checks whether the current station service is available (Line 3). If the transfer station service is unavailable, check whether the service recovery time is acceptable (Line 4 and Line 10). If waiting time for the service does not result in an order timeout, the algorithm returns the plan with less energy consumption (Lines 5-9). On the contrary, the system will re-route and find a suitable transfer station instead (Lines 11-12). The time complexity of Algorithm 2 mainly comes from the shortest path calculation between UGV stations near the user in the second half of the order. Assuming that the number of UGV stations set by the logistics company near the user is m, the time complexity of Algorithm 2 is $O(m^2)$.

[2]https://lbs.amap.com/

Algorithm 2 Energy-Aware Dynamic Service Resolution Algorithm (EDSR)

Input: Delivery Order Request R, DaaS$^+$ service model diagram D_k, delivery service provisioning plan DS, GS service restoration time T_{wait};
Output: P_{res}; ▷ Optimal service resolution plan;

1: **procedure**
2: $RS \leftarrow \varnothing$
3: **if** $D_k.GS_m$ is Not Available **then**
4: **if** $DS.T_{sum} + T_{wait} \geq R.D$ **then**
5: $GSList = getUseableList(GS_m)$
6: $RS \leftarrow getRouteByDijkstra(GSList)$
7: **if** $RS.ER_{sum} > DS.E_{sum}$ **then**
8: $RS \leftarrow \varnothing$
9: **end if**
10: **else if** $DS.T_{sum} + T_{wait} > R.D$ **then**
11: $GSList = getUseableList(GS_m)$
12: $RS \leftarrow getRouteByDijkstra(GSList)$
13: **end if**
14: **end if**
15: **return** RS
16: **end procedure**

TABLE III
SETTING OF THE DELIVERY ORDER

Order ID	Start Node	Destination Node	Cargo Weight/g	Deadline/s
1	A_1	C_1	500-2000	1500
2	A_4	C_3	500-2000	1500
3	A_5	C_2	500-2000	1500
4	A_9	C_2	500-2000	1500
5	A_{10}	C_1	500-2000	1500
6	A_{11}	C_3	500-2000	1500

V. SIMULATION AND EXPERIMENT

A. Parameter Setting and Experiment Platform

In this section, we conduct comprehensive simulation experiments using real-world UAV-UGV integrated network dataset as shown in Fig. 5.[3] It is with the six customer nodes C_i, twenty-two UAV delivery stations A_i, seventeen UGV delivery stations G_i. Each of the nodes is connected by a pre-set UAV airline or an unmanned vehicle path. The start and destination of the delivery order may be any pair of these nodes. Table III illustrates the setup of the delivery orders, including the start node, destination node, cargo weight, and deadline [17]. Taking into account the average distance covered by delivery routes and the typical speed of delivery equipment, a standardized delivery order deadline of 25 minutes is set. The proposed ES-PSO mechanisms are implemented by using JAVA and all experiments are run based on EXPRESS 2.0.[4] EXPRESS 2.0 is an intelligent service management framework for AIoT in the edge [12].

The setting of the UAV-UGV integrated network environment is listed in Table IV. Each UAV or UGV delivery station is equipped with one of the three different types of delivery

[3]Experiment dataset and source code: https://github.com/ISEC-AHU/ES-PSO
[4]EXPRESS 2.0 platform: https://github.com/ISEC-AHU/EXPRESS2.0

TABLE IV
SETTING OF LAST-MILE DELIVERY ENVIRONMENT

Variable	Value
Number of nodes	45
Number of UAVs at UAV station	3
Number of UGVs at UGV station	3
UAV avgFlySpeed (m/s)	5, 6, 7
UAV Load (kg)	1.5, 3, 6
UAV batCapacity (Ah)	3, 4.5, 6
UAV noLoadPower (w)	200, 300, 500
UAV max power (w)	300, 400, 600
UGV Speed (m/s)	2, 3, 4
UGV Load (kg)	10, 15, 20
UGV batCapacity (Ah)	12, 15, 18
UGV noLoadPower (w)	30, 50, 80
UGV max power (w)	80, 100, 130

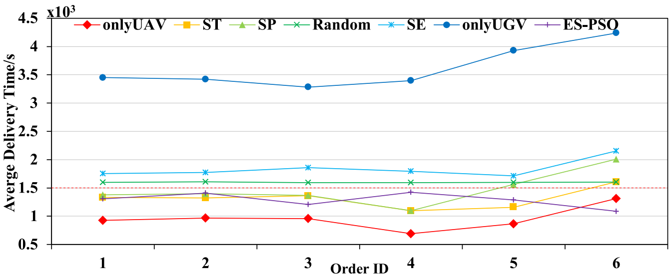


Fig. 6. Evaluation of delivery time for single delivery order.

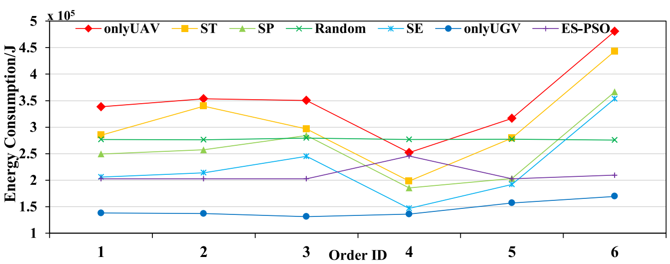


Fig. 7. Evaluation of delivery energy consumption for single delivery order.

equipment. To denote the validity of ES-PSO, we select six methods, i.e., OnlyUAV [29], OnlyUGV [10], Random, Shortest Path (SP) [30], Shortest Time (ST) [31] and Shortest Energy (SE) [32]. To avoid experimental bias, all results are averaged after 50 runs as the final results.

B. Delivery Energy Consumption

In this subsection, our attention is directed towards assessing the performance of the ES-PSO mechanism. The simulation results for single delivery order, including delivery time and energy consumption with varying cargo weights, are presented in Figs. 6 and 7. Further, the simulation outcomes for multiple delivery orders, encompassing average time and total energy consumption, are depicted in Figs. 8 and 9.

1) Single Delivery Order: The findings from Fig. 6 illustrate that, for a single order with varying cargo weights, the ES-PSO mechanism consistently yields the optimal service provisioning plan that meets the time constraint (depicted by the red line indicating the order delivery deadline). Additionally, Fig. 7 demonstrates that the ES-PSO mechanism generates a service provisioning plan with the lowest energy consumption among all plans adhering to the deadline constraint when compared to alternative methods. Specifically, the energy consumption of the ES-PSO mechanism is 40.18%, 28.89%, 18.78%, and 26.82% lower than that of onlyUAV, ST, SP, and Random methods, which may meet the deadline.

Fig. 8. Evaluation of delivery time for multiple delivery orders.

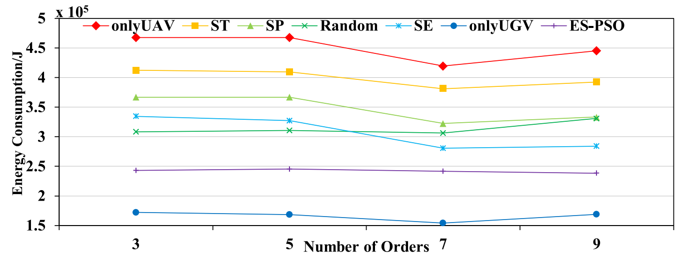


Fig. 9. Evaluation of delivery energy consumption for multiple delivery orders.

2) Multiple Delivery Orders: As depicted in Fig. 8, in the case of multiple orders with random cargo weights, the ES-PSO mechanism provides a service provisioning plan that always satisfies the deadline constraint (the red line). Fig. 9 show that ES-PSO mechanism also achieves the lowest total energy consumption among all methods that satisfy the deadline constraint. For instance, when the number of order is 3, the total energy consumption of the ES-PSO mechanism is 47.96%, 40.98%, 33.69% lower than onlyUAV, ST, SP methods. According to the experimental results in Figs. 6 to 9, even though some methods generate service provisioning plan that have lower energy consumption than ES-PSO, a comparison of the delivery time shows that the order completion times exceed the deadline constraints (e.g., onlyUGV).

Therefore, it is evident that ES-PSO mechanism is more effective than other baseline algorithms in optimizing both time and energy consumption in the static service phase. This is primarily because the ES-PSO mechanism has taken into account the deadline constraint, the weight of the cargo, and the energy consumption of the UAV and UGV. The ES-PSO mechanism identifies the optimal service provisioning plan with the least energy consumption, while adhering to the delivery order's deadline constraint.

C. The Impact of Penalty Coefficient

The penalty coefficient is a representative of the rate of time by which delivery orders exceed their deadline constraint. Hence, the crucial issue is to establish penalty coefficients in a manner that enables the FIUV function to accurately evaluate service provisioning plans.

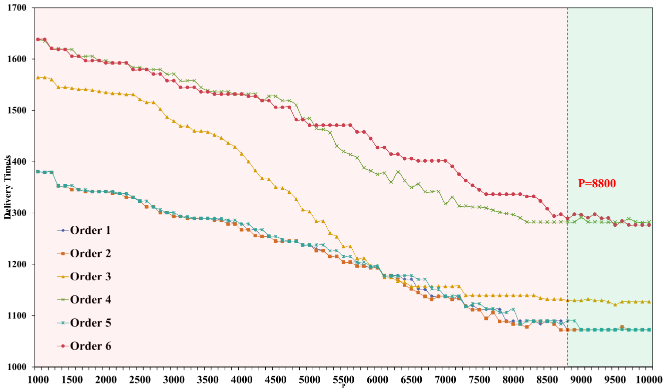


Fig. 10. Evaluation of penalty coefficient for single delivery orders 1 to 6.

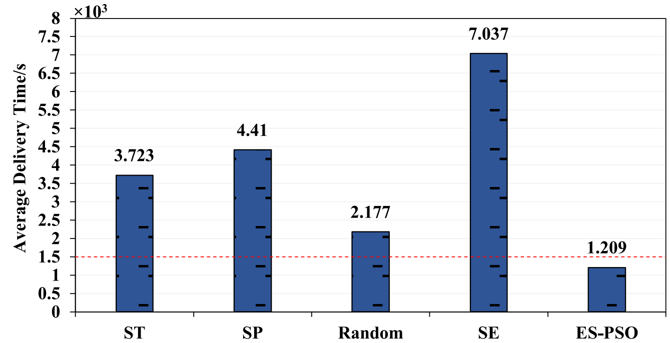
Fig. 10 shows the evaluation of penalty coefficient for single delivery orders 1 to 6. It is evident that as the penalty coefficient grows, the delivery time of all the delivery service provisioning plans generated by the ES-PSO mechanism keeps decreasing. For example, it is clear from this figure the ES-PSO with $P = 8800$ has the best performance in terms of the lowest delivery time for all orders 1 to 6.

D. Resolution of Service Unavailability

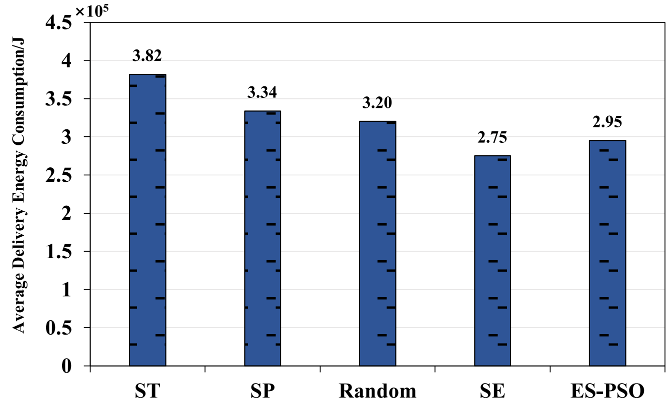
When a service unavailable arises during the delivery process, prompt resolution is imperative to ensure on-time delivery of orders. Before the delivery process initiates, the ES-PSO mechanism independently formulates optimal delivery service provisioning plans for distinct delivery orders in the static phase. During the actual delivery process, potential conflicts arising from the use of different delivery services for various orders may lead to instances of service unavailability. Hence, the frequency of service unavailability is determined based on the conflicts arising from the optimal service provisioning plans generated in the static phase.

In this subsection, we delve into the assessment of the dynamic stage's service unavailability resolution. The results illustrated in Fig. 11(a) underscore that the proposed ES-PSO mechanism adeptly manages service unavailable in the delivery process, guaranteeing timely arrivals for all orders. Conversely, baseline algorithms encounter varying degrees of delivery delays, contributing to a notable increase in average delivery time. The ES-PSO mechanism demonstrates a remarkable 80% to 482% reduction in average order delivery time compared to baseline algorithms.

Furthermore, Fig. 11(b) depicts that the average delivery energy consumption of the delivery service provisioning plan generated by our ES-PSO mechanism which is only approximately 7.4% higher than that of the lowest energy consumption algorithm. This marginal increase in energy consumption is deemed necessary for rerouting the order delivery path to address service unavailable. In essence, the proposed mechanism effectively navigates service unavailable, minimizing delivery delays without incurring a substantial increase in energy consumption.



(a) Average delivery time



(b) Average delivery energy consumption

Fig. 11. Service Unavailability Effects on Delivery Time and Energy Consumption.

E. Impact of Cargo Weight

The main optimization objective of our algorithm is the total energy consumption of cargo delivery. Fig. 12 shows that the energy consumption growth rate of our algorithm decreases from 42% to 80% compared to other algorithms as the weight of the cargo increases, while all cargoes are guaranteed to be delivered on time. This is because our algorithm optimizes the selection of delivery devices, not simply based on the cargo weight, but by selecting the globally optimal type of delivery devices.

F. Ablation Study

As mentioned in Section IV, the ES-PSO mechanism consists of a series of sub-algorithms, namely SPSO, EDSR, and DCS. In order to ascertain the indispensability of these sub-algorithms, we have devised a set of baseline algorithms for verification. As depicted in Table V, a comprehensive set of seven baseline algorithms has been established, each formed by different combinations of sub-algorithms, in addition to the ES-PSO mechanism.

Fig.s 13 and 14 underscore the pivotal role played by the three sub-algorithms embedded in our optimization algorithm in enhancing the effectiveness of delivery energy consumption. The DCS algorithm optimizes the delivery service selection plan, and its absence leads to an approximate 4% increase in average delivery time and a 15% increase in average energy

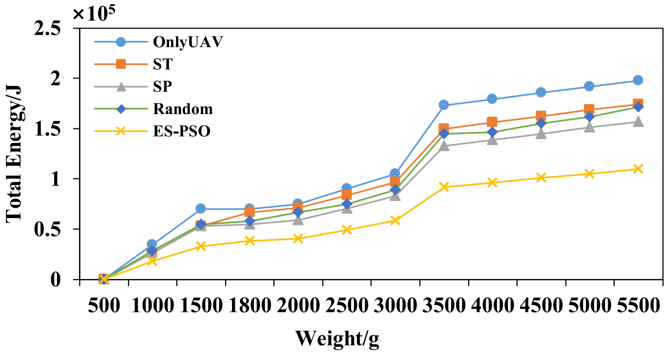


Fig. 12. Impact of cargo weight.

TABLE V
SETTING OF THE BASELINE ALGORITHM

Baseline Algorithm	SPSO	EDSR	DCS
SP	0	0	0
DCS	0	0	1
EDSR	0	1	0
EDSR-DCS	0	1	1
SPSO	1	0	0
SPSO-DCS	1	0	1
SPSO-EDSR	1	1	0
ES-PSO	1	1	1

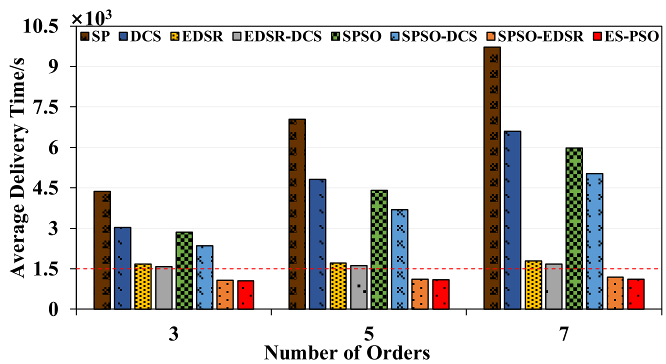


Fig. 13. Average delivery time for ablation study.

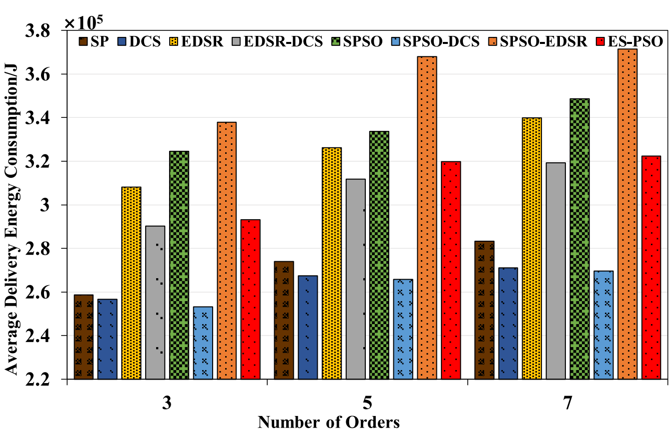


Fig. 14. Average energy consumption for ablation study.

consumption. This discrepancy arises because, near the maximum load of the delivery devices, our energy consumption model indicates that their power at runtime approaches its maximum. Substituting a larger delivery device at this juncture may be advantageous, as it not only accelerates delivery but also reduces total energy consumption due to the decrease in delivery time.

The SPSO algorithm is instrumental in determining the delivery service provisioning plan that minimizes energy consumption for delivery within deadline constraints. Without the intervention of the SPSO algorithm, there is no assurance that all orders can be delivered on time. The EDSR algorithm addresses service unavailable during order delivery by sacrificing some energy consumption to guarantee the timely arrival of all orders. The absence of the EDSR algorithm would result in a significant number of orders failing to meet delivery deadlines due to backlog.

In essence, the collaboration of all three sub-algorithms is imperative for optimization of delivery energy consumption while ensuring on-time delivery of cargoes.

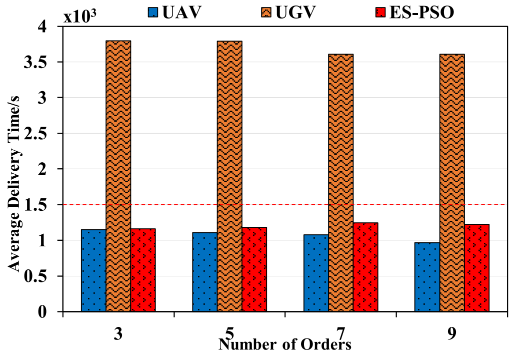
G. Comparison of Different Delivery Solutions

To demonstrate the effectiveness of the proposed DaaS$^+$ architecture and the ES-PSO algorithm in this paper, we have compared two mainstream automated delivery solutions: UAV delivery and UGV delivery.

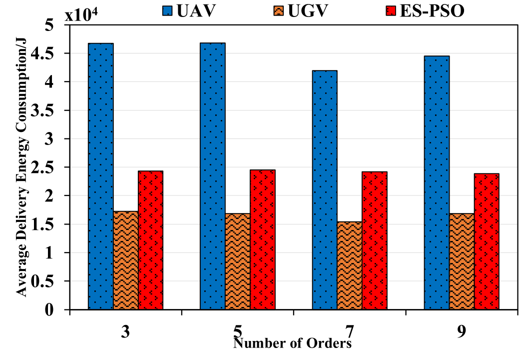
As shown in Fig. 15, the results indicate that the UAV delivery mode exhibits advantages in terms of delivery time metrics but performs poorly in energy consumption metrics (exceeding the other two solutions by 63.11% and 46.06% respectively). Conversely, the UGV delivery mode demonstrates advantages in delivery energy consumption metrics but performs less favorably in delivery time metrics (exceeding the other two solutions by 70.97% and 67.45% respectively). Only the ES-PSO algorithm under the DaaS$^+$ architecture proposed in this paper achieves the goal of minimizing energy consumption while meeting delivery deadlines (delivery energy consumption lower than the UAV mode by 46.06%). This is attributed to the ability of the proposed model to leverage the distinct attributes of UAV and UGV delivery services based on the characteristics of delivery tasks and allocate appropriate delivery services reasonably according to delivery requests.

H. Real-World Experiment Demonstration

After conducting the aforementioned simulation experiments, it is imperative to validate the performance of the ES-PSO mechanism in a real-world delivery scenario. In this regard, we utilize actual delivery samples from the EXPRESS 2.0 platform, featuring a $8km \times 5km$ delivery network map encompassing twenty-two UAV delivery stations, seventeen UGV delivery stations, and one delivery destination. The delivery order's start node and destination node is W_1 and C_1. The cargo weight is $1kg$. The deadline constraint is $1500s$. ES-PSO, along with three other comparative algorithms (ST, SP, SE), are employed in this scenario to generate an optimal delivery service provisioning plan for the UAV-UGV integrated network. The decision to exclude random algorithm is attributed to their limited suitability within practical systems. Random algorithm often lacks the deterministic and adaptive characteristics necessary for optimizing the complex dynamics and specific constraints present in real-world delivery

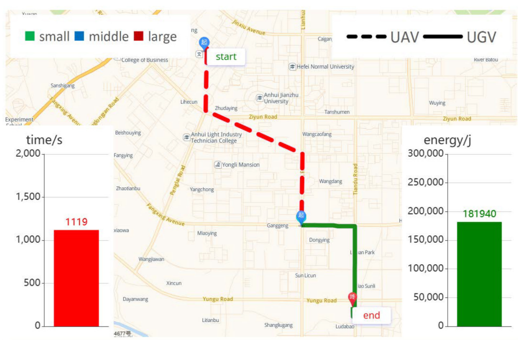


(a) Average delivery time

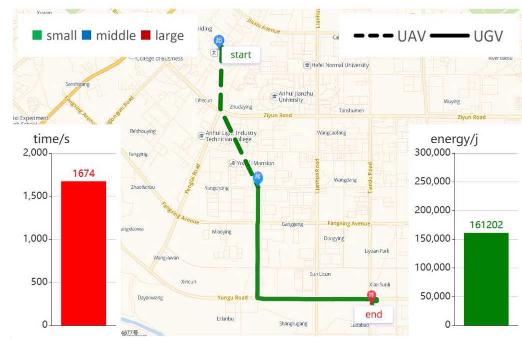


(b) Average delivery energy consumption

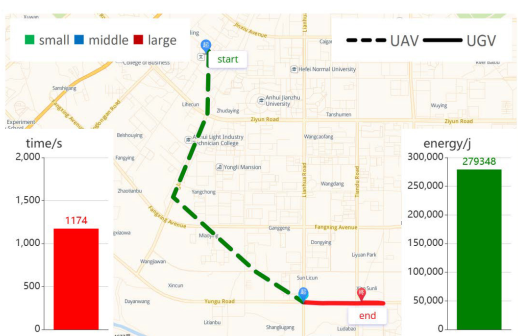
Fig. 15. Comparison among different delivery modes.



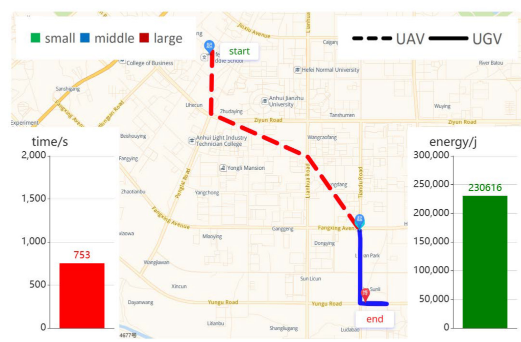
(a) ES-PSO mechanism



(b) SE algorithm



(c) SP algorithm



(d) ST algorithm

Fig. 16. Evaluation of delivery service provisioning plans in EXPRESS 2.0 platform. DEMO video: https://youtu.be/GHKD_VvJD88.

operations. Consequently, deterministic algorithms like ES-PSO, ST, SP, and SE were chosen for their structured and systematic approach, better suited to address the intricacies and constraints inherent in the context of UAV-UGV integrated network delivery services.

The delivery service provisioning plans generated by the four algorithms are presented in Fig. 16. The left side of each sub-figure illustrates the timeline of the delivery service scenario, while the right side displays the energy consumption associated with the scenario. The delivery paths in the figure are color-coded, with green, blue, and red representing small, medium, and large delivery UAV/UGV, respectively. Dashed lines indicate UAV delivery paths, while solid lines represent UGV delivery paths. It is evident that, under the imposed deadline constraint (the SE algorithm exceeds the deadline constraints), the ES-PSO mechanism achieves a remarkable 34.87%, 21.11% reduction in delivery energy consumption when compared to the SP and ST algorithms. This experiment thus substantiates the effectiveness of ES-PSO mechanism when applied in practical systems.

VI. Conclusion and Future Work

We expanded the existing delivery network Drone-as-a-Service (DaaS) to develop a UAV-UGV integrated network architecture based on Multi-access Edge Computing (MEC), referred to as DaaS^{+}. Within this architecture, we formulated an Energy-Aware Holistic Service Provisioning Mechanism based on Particle Swarm Optimization (ES-PSO), aimed at optimizing energy consumption under deadline constraints for the UAV-UGV integrated network. The experimental results convincingly showcased the superiority of our proposed mechanism when compared to existing baseline approaches.

Our future research will delve into exploring the influence of environmental uncertainties, such as weather and network conditions, intending to provide a more comprehensive evaluation and enhance the robustness of our UAV-UGV integrated network architecture in real-world scenarios. In addition, the challenges faced by UAV-UGV integrated delivery systems in real-world scenarios, including regulation, cost, and scalability aspects also deserve more in-depth study.

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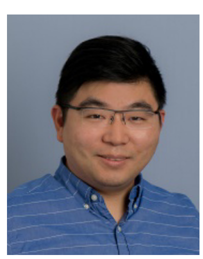

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